

Operators manual for Attachment



Manufacturer:

AVANT[®]
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
FIN-33470 YLÖJÄRVI
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

www.avanttecno.com

1. Introduction

Avant Tecno Oy wants to thank you for purchasing this original Avant attachment. It is the result of Avant's long experience in design and manufacturing of attachments. We ask you that you read and understand the contents of this manual completely before operating the attachment. This will improve your operating and maintenance efficiency, help avoid breakdowns and damage and extend your machines life. Do not hesitate to contact your local Avant dealer for advice regarding service, parts and any problems relating to the operation of your attachment.

2. Designed purpose

Avant snow blower is designed for Avant 200- and 500-series loaders only. It is designed for removing snow from asphalt or other solid surfaces. The snow blower needs at least 10 cm of fallen snow for it to start working properly. Less than 10 cm snow might not fly far enough, because the snow blower cannot create enough mass to the snow.

3. Safety instructions



- Never use the snow blower without instructions. Read all the decals and this manual.
- Start the operation slowly and carefully.
- Do not wear loose clothing, long uncovered hair or jewellery near the machine.
- When coupling the attachment, make sure that the attachment locking pins lock in positively.
- Keep hands, feet and clothing away from any moving part.
- Make sure, that there are no bystanders in the working area. **The safety area is 35 meters or 100 feet.**
- Before starting the blower make sure, that the outlet discharge chute of the snow blower is turned to a safe direction.
- Do not leave the cockpit while the snow blower is running.
- Always keep the switch of the discharge chute rotation on your hand.
- Never go under an uplifted attachment.
- Never turn the discharge chute beyond the foremost 180 degrees. The driver might result in serious injury if the discharge chute is pointed towards the cabin.
- Always carry the attachment as low as possible.
- Never perform any service operations when the engine is running.

4. Technical specification

Operating width:	1100 mm
Total width:	1220 mm
Total height:	1200 mm
Total length:	750 mm
Weight:	110 kg
Throwing distance:	8 – 12 m

- The snow blower is equipped with a 12V electric motor, which rotates the outlet discharge chute.

5. Attaching the snow blower to the loader

5.1 Mechanical attachment of the snow blower

The snow blower is attached to the loader with the loaders quick attachment plate and the counter part on the snow blower.

Coupling of the attachment into the attachment coupling plate happens as follows:

Stage 1:

- Lift up the two locking pins on either side of the attachment coupling plate and turn them backward so that they remain in the locked up position.
- Make sure, that the pins remain in the up position; otherwise you cannot couple the attachment properly!



Stage 2:

- Turn the attachment coupling plate with the tilt movement so that the upper edge of the plate leans forward.
- Drive the loader into the attachment.



Stage 3:

- Lift the boom a little so that the attachment lifts off of the ground.
- Pull the boom control lever to the left so that the lower edge of the attachment coupling plate turns into the attachment.



Stage 4:

- Pull the boom control lever more to the left so that attachment coupling plate turns more and the bolts on the boom push the pins down in the holes of the attachment.
- **CAUTION! Make sure that the pins lock properly down in the holes of the attachment.**



Always keep in mind the safety instructions from paragraph 3, when attaching the snow blower.

5.2 Connecting the hydraulic couplings

Coupling the 500 series loader

The snow blower has three hydraulic quick couplings, which should be attached to the corresponding couplings in the loader. There are two male couplings and one female coupling in the snow blower. The loader has correspondingly one male coupling and two female couplings.

The hoses from the snow blower should be attached to the loader correctly (pressure line to pressure line and return line to return line). The blue rubber caps of the couplings always mark the pressure line (one on the loader and one on the attachment). Always connect them as a pair. This way the snow blower starts to rotate in the right direction, when the loader's auxiliary hydraulics control lever is in its locked position.



If the snow blower's auger does not start to rotate when the loader's auxiliary hydraulics control lever is turned to right in its locked position, please change the coupling of the return- and pressure hose. **See picture 1 in the next page**



Picture 1



Always **make sure**, that **all couplings are properly locked in before starting the engine**. The hydraulic motor of the snow blower may get damaged, if one of the hoses is not locked in.

- For coupling the hydraulic hoses see also the operators manual of the loader.
- Always remember the safety instructions from paragraph 3, when coupling the attachments.

Coupling the 200 series loader

The snow blower has three hydraulic quick couplings, which should be attached to the corresponding couplings in the loader. There are two male couplings and one female coupling in the snow blower. The loader has correspondingly one male coupling and two female couplings.

The hoses from the snow blower should be attached to the loader correctly (pressure line to pressure line and return line to return line). The red caps of the couplings always mark the pressure line (one on the loader and one on the attachment). This way the snow blower starts to rotate in the right direction, when the loaders auxiliary hydraulics pedal is pressed down to its locked position.

The 200 series machines are equipped with flat face couplings, because they are more convenient and clean in everyday use. However there are some things you have to remember when coupling and uncoupling a flat face coupling.

1. Always try to **keep the couplings as clean as possible**. Dirt, ice etc might prevent the coupling from functioning properly.
2. Although the joint surface seems flat there are sprung parts in the coupling, which give in when pressed together correctly. If the couplings do not lock in to each other try **to release pressure by moving the auxiliary hydraulics pedal up and down** a couple of times and then try again.
3. **Before uncoupling the couplings always stop the engine** from of the loader and **move the auxiliary hydraulics pedal up and down** a couple of times. This releases any pressure left to the attachment and ensures easy coupling the next time you are using the attachment.

4. Flat face couplings have a locking mechanism, which prevents them to uncouple accidentally. When removing them from each other make sure that the ball and the slot from the female coupling are facing each other. **See picture 2**



Picture 2



Always **make sure**, that **all couplings are properly locked in before starting the engine**. The hydraulic motor of the snow blower may get damaged, if one of the couplings is not locked in.



If the female coupling will not release from the male coupling when its collar is pulled backwards, rotate the collar until the slot in it is facing the ball (**see picture 2**) and try again.



If the snow blower does not start to rotate when the loader's auxiliary hydraulics pedal is pressed down to its locked position, please change the coupling of the return- and pressure hose. **See picture 3**



Picture 3

- For coupling the hydraulic hoses see also the operators manual of the loader.
- Always remember the safety instructions from paragraph 3, when coupling the attachments.

5.3 Connecting the electric wire for the discharge chute rotation

The discharge chute rotation is operated by an electric toggle switch. This switch is attached to the wire coming from the root of the discharge chute. In the other end of the wire is a plug, which fits in to the 12V outlet on the loaders dashboard. (If you cannot find the outlet; see the operators manual of the loader) When the plug is connected to the connector and power is on in the loader you are able to rotate the discharge chute.

The switch works in two directions. When it is turned to the other side it moves the discharge chute to another direction and vice versa. When you release the switch the rotation stops automatically. **See picture 4**



Picture 4



Always keep the switch in your hand while operating the snow blower. This way you are able to change the discharge chute angle rapidly, if needed.



Do not point the discharge chute towards people or property. The snow blower might throw out some small stones or other small objects. The **safety area is 35 meters or 100 feet**.



Never turn the discharge chute so that it points towards the operator. **Keep the discharge chute** in a position where it points **to the side** (left or right) **or straight forward**, but **NEVER towards the operator**. The driver might result in serious injury if the discharge chute is pointed towards the driver.

6. Working with the snow blower

After the snow blower is attached to the loader's quick attachment plate, the hydraulic couplings are attached and the electrical wire for the discharge chute rotation is plugged in you are ready for work. Before turning the auger on, make sure that the discharge chute is pointed to a safe place. Turning the auxiliary hydraulics lever right to its lock position activates the auger. (In 200 series press the auxiliary hydraulics pedal down to its lock position)

When you are working with the snow blower you should keep the engine revolutions at full in order to get the best possible ejection. If you have a plus-model loader (520+, 528+ or 530+) the snow blower works best with both pumps. (2-Pump selection on the loader's right side control panel, see loader's operators manual)

7. Service

Larger overhauls and possible repair works should always be done by an authorized Avant service. Contact your local Avant dealer for the nearest service if you need assistance.

The operator can do the greasing himself. In the following are few pointers how to do this overhaul procedure.

7.1 Greasing

For greasing you need a grease press and a tube of grease, which will fit to the press. Both of these are available in a regular hardware store.

The snow blower has three (3) greasing points. One in the root of the discharge chute and two in the auger bearings. **See picture 5**



Picture 5

The auger bearings should be greased after every 7-10 working hours and the discharge chute rotator should be greased approximately twice a year. Do not add more than a couple of presses from the grease press per one overhaul.

By following these greasing instructions you will increase the life of your snow blower significantly.

7. Warranty terms for Avant snow blower

(Product number A2847 for 500 series)

(Product number A21097 for 200 series)

During the first year of operation Avant Tecno Oy warrants to replace any part or repair any defect, which may occur, subject to the terms below:

1. The product has received regular maintenance in accordance with schedules given by the manufacturer.
2. Any damage caused by operation in a negligent manner or exceeding the approved specifications detailed in this manual is excluded.
3. Avant Tecno Oy accepts no responsibility for interruption to working or any other consequential losses resulting from any failure of the product.
4. Only Avant Tecno Oy approved replacement or original quality parts shall be used during routine maintenance.
5. The Avant Warranty excludes any consumable parts (e.g. chains or bearings) except where it can be clearly shown that these parts were defective on original supply.

In the event a fault occurs which is attributable to manufacturing or assembly defect you should arrange to return your attachment to your authorized dealer for repair. Travel and freight costs are excluded.